Alluxio AWS c5n.9xlarge (8 clients) + i3en.12xlarge (8 storage)

MLPerf Storage v2.0 Submission

System Name: AWS c5n.9xlarge (8 clients) + i3en.12xlarge (8 storage)

Submitter Name: Alluxio Storage

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Benchmark Version: v2.0

Solution Type: Alluxio (Distributed Filesystem with AWS S3)

Submission Division: CLOSED

System Topology Diagram:

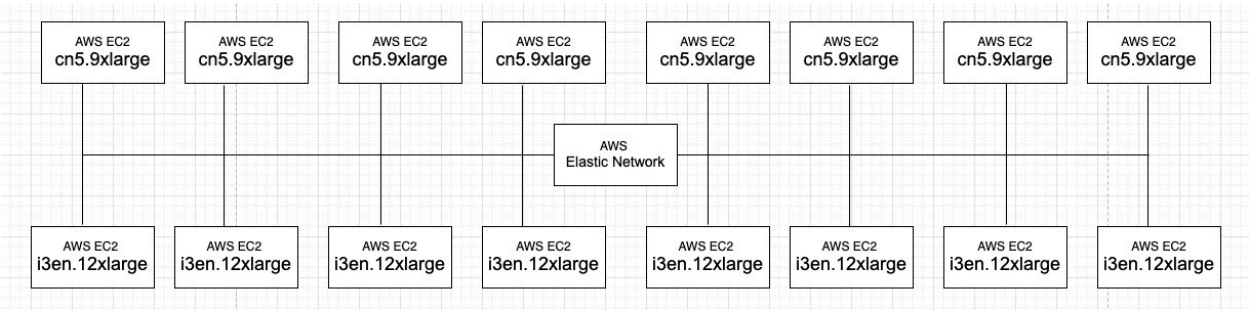


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1 Hardware

The system comprises 8 AWS EC2 c5n.9xlarge instances as client and 8 i3en.12xlarge instances as storage nodes, interconnected via AWS Elastic Network Adapter (ENA) for high-speed networking. The total system includes four nodes, with a combined rack unit estimate of approximately 8U based on typical server configurations for these instance types.

1.1 Client

Each client is an AWS EC2 c5n.9xlarge instance (compute-optimized) with:

- Processor: 1x Intel Xeon Platinum 8259CL, 48 cores
- Memory: 96 GiB RAM
- Network: Elastic Network Adapter (ENA), up to 50 Gbps
- Storage: No local storage (ephemeral storage not used)
- Accelerators: None
- Rack Units: Estimated 2U per node (total 4U)

The clients communicate with the storage nodes via ENA, accessing the Alluxio distributed filesystem mounted at /mnt/alluxio-fuse.

1.2 Storage

Each storage node is an AWS EC2 i3en.12xlarge instance, configured as:

- Processor: 1x Intel Xeon Platinum 8259CL, 48 cores
- Memory: 384 GiB RAM
- Storage: NVMe SSD (ephemeral), 4x 7.5 TiB per node (total 30 TiB)
- Accelerators: Up to 64x NVIDIA H100 per node (total up to 128 H100 accelerators)
- Network: Elastic Network Adapter (ENA), up to 75 Gbps
- Rack Units: Estimated 2U per node (total 4U)

The storage nodes host the Alluxio 3.6.0 distributed filesystem, with AWS S3 as the underlying storage layer, providing high-performance storage to the clients.

1.3 Network

The clients and storage nodes are connected via AWS Elastic Network Adapter (ENA), providing high-speed, low-latency networking within the AWS datacenter. No physical switch details are applicable as this is a cloud-based system.

1.4 Power Requirements

The power requirements for AWS EC2 instances are managed by AWS datacenter infrastructure. Exact PSU details are not publicly available, but estimated power consumption is provided based on typical server configurations for similar instance types. The system uses AWS datacenter standard cooling.

Component	Nameplate Rated Power	Design Power
c5n.9xlarge (8 Clients)	6400 W (est.)	4800 W (est.)
i3en.12xlarge (8 Storage)	9600 W (est.)	8000 W (est.)
Total	**16,000 W (est.)**	**12,800 W (est.)**

2 Software

2.1 Client

Each client runs:

- Operating System: Ubuntu 22.04
- Software Stack: Alluxio 3.6.0 FUSE client, OpenJDK 11, FUSE, Docker, MLPerf Storage benchmark code (v2.0-rc0, cloned from https://github.com/mlcommons/storage.git)

The Alluxio FUSE client is used to mount the distributed filesystem at /mnt/alluxio-fuse, accessing data stored on AWS S3 via the Alluxio namespace /mlperf_data.

2.2 Storage

Each storage node runs:

- Operating System: Ubuntu 22.04

- Software Stack: Alluxio 3.6.0 server, OpenJDK 11, FUSE, Docker

The Alluxio server manages the NVMe SSD storage (ephemeral) and integrates with AWS S3 as the underlying storage layer, providing the distributed filesystem to the clients.

2.3 Network

No specific network software configuration is required beyond the default AWS ENA drivers included in Ubuntu 22.04.

3 Settings

3.1 Client

The Alluxio 3.6.0 FUSE client is installed using the following setup script (setup.sh):

```
#!/bin/bash

sudo apt update

sudo apt install -y openjdk-11-jdk fuse curl wget

wget
https://downloads.alluxio.io/downloads/files/3.6.0/alluxio-3.6.0-bin.tar.gz

tar -xzf alluxio-3.6.0-bin.tar.gz
```

The Alluxio client configuration file (alluxio-site.properties) is generated automatically with:

- alluxio.user.file.writetype.default=CACHE_THROUGH
- alluxio.user.block.size.bytes.default=64MB
- alluxio.user.network.netty.timeout.ms=10000

The MLPerf Storage benchmark code is installed by cloning:

```
git clone -b v2.0-rc0 https://github.com/mlcommons/storage.git
```

No additional tuning was applied, as permitted in the CLOSED division.

3.2 Storage

The Alluxio 3.6.0 server is installed using the same setup script as the client. The initialization script (init_datasets.sh) configures the Alluxio filesystem:

```
#!/bin/bash

set -e

alluxio-start.sh local

alluxio fs mount /s3bucket s3://your-bucket-name

mkdir -p /mnt/alluxio-fuse

alluxio-fuse mount /mnt/alluxio-fuse
```

No tuning was applied beyond the default configuration.

3.3 Network

No specific network tuning was applied, relying on AWS ENA defaults.

4 Miscellaneous

4.1 Drop Caches

Caches on both client and storage nodes were cleared between iterations using:

```
#!/bin/bash

sync

echo 3 > /proc/sys/vm/drop_caches
```

This was executed on all nodes to ensure consistent benchmarking conditions.

4.2 Benchmark Execution

The benchmark is executed using the following steps:

1. Mount Alluxio:

```
alluxio-fuse mount /mnt/alluxio-fuse
```

2. Generate Test Data:

```
mlpstorage training datasize -m resnet50 --client-host-memory-in-gb 100
--max-accelerators 15 --num-client-hosts 1 --accelerator-type h100
--results-dir /home/ubuntu/mlps_results

mlpstorage training datagen --hosts=127.0.0.1 --model=resnet50 --exec-type=mpi
--param dataset.num_files_train=3742 --num-processes=30
--results-dir=/home/ubuntu/mlps_results
--data-dir=/mnt/alluxio/alluxio-fuse/s3/resnet50_3742_499GB
```

3. Run Benchmark Test:

```
mlpstorage training run --hosts 127.0.0.1 --num-client-hosts 1
--client-host-memory-in-gb 100 --num-accelerators 15 --accelerator-type H100
--model resnet50 --data-dir /mnt/alluxio/alluxio-fuse/s3/resnet50_3742_499GB
--results-dir /home/ubuntu/mlps_results --param dataset.num_files_train=3742
--allow-run-as-root --oversubscribe
```

4. Collect Results:

   Results are collected from /home/ubuntu/mlps_results and converted into result_*.txt format for submission.*

The run script uses the following bash options for robustness:

```
#!/bin/bash

set -u

set -e
```

```
set -x
```

```
set -o pipefail
```

No modifications were made to the reference MLPerf code, ensuring compliance with CLOSED division requirements.

5 Rack Unit Totals

The estimated rack unit totals for the system are:

- *Clients (8x c5n.9xlarge): 4U (est.)*
- *Storage (8x i3en.12xlarge): 4U (est.)*
- *Total: 8U (est.)*

Note: Rack unit estimates are based on typical server hardware configurations, as AWS EC2 instances are virtualized.